Monitoring in Anesthesia MCQ – Chapter 6

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 3: Quintessential in Anesthesia (Chapter 6)

50 Questions • 30 Minutes

Instructions:
• Each question has 4 options (A, B, C, D).
• The correct answer is shown on the right side of each question.
• Detailed explanations with textbook page references are at the end.

Questions

Q1. Which of the following is a method of clinical monitoring during anesthesia?

- **A.** Pulse oximetry
- **B.** Observation of skin color, pupil size, and respiratory pattern
- **C.** Invasive arterial pressure
- **D.** BIS monitoring

Ans: B

Q2. Clinical signs of adequate depth of anesthesia include all EXCEPT:

- **A.** Stable heart rate and blood pressure
- **B.** Absence of sweating and lacrimation
- **C.** BIS value of 40–60
- **D.** Central and constricted pupils

Ans: C

Q3. Dilated pupils during anesthesia may indicate:

- **A.** Adequate depth of anesthesia
- **B.** Light anesthesia, hypoxia, or hypercarbia
- **C.** Excessive muscle relaxation
- **D.** Hypothermia

Ans: B

Q4. The minimum mandatory monitoring standards during anesthesia include all EXCEPT:

- **A.** Pulse oximetry
- **B.** ECG
- **C.** Capnography
- **D.** Cardiac output monitoring

Ans: D

Q5. ECG monitoring during anesthesia primarily detects:

- **A.** Cardiac output
- **B.** Arrhythmias, ischemia, and heart rate
- **C.** Blood pressure
- **D.** Venous return

Ans: B

Q6. Which ECG lead is most sensitive for detecting intraoperative myocardial ischemia?

- **A.** Lead I
- **B.** Lead II
- **C.** Lead V5
- **D.** Lead aVR

Ans: C

Q7. Lead II on ECG is preferred during anesthesia for monitoring:

- **A.** Left ventricular ischemia
- **B.** P waves and inferior wall ischemia
- **C.** Right bundle branch block
- **D.** Pericardial effusion

Ans: B

Q8. Non-invasive blood pressure (NIBP) measurement by oscillometric method measures: **A.** Only systolic pressure directly **B.** Mean arterial pressure directly, then calculates systolic and diastolic **C.** Only diastolic pressure directly **D.** Cardiac output	**Ans: B**
Q9. Invasive arterial blood pressure monitoring is indicated in all EXCEPT: **A.** Major cardiovascular surgery **B.** Patients requiring frequent arterial blood gas analysis **C.** Minor ambulatory surgery in healthy patients **D.** Patients with severe hemodynamic instability	**Ans: C**
Q10. The most common site for invasive arterial cannulation is the: **A.** Brachial artery **B.** Radial artery **C.** Femoral artery **D.** Dorsalis pedis artery	**Ans: B**
Q11. Allen's test is performed before radial artery cannulation to assess: **A.** Venous drainage of the hand **B.** Adequacy of collateral circulation from the ulnar artery **C.** Nerve supply to the hand **D.** Patency of the brachial artery	**Ans: B**
Q12. Central venous pressure (CVP) monitoring catheter tip should ideally be positioned at: **A.** Right atrium **B.** Junction of superior vena cava and right atrium **C.** Right ventricle **D.** Pulmonary artery	**Ans: B**
Q13. Normal CVP value is: **A.** 0–2 mmHg **B.** 2–8 mmHg (3–11 cm H_2O) **C.** 15–20 mmHg **D.** 25–30 mmHg	**Ans: B**
Q14. CVP monitoring is useful for assessing: **A.** Left ventricular function **B.** Right heart filling pressure and intravascular volume status **C.** Pulmonary artery pressure **D.** Cerebral perfusion pressure	**Ans: B**

Q15. The pulmonary artery catheter (Swan-Ganz catheter) can measure all EXCEPT: **A.** Pulmonary artery pressure **B.** Cardiac output by thermodilution **C.** Pulmonary capillary wedge pressure **D.** Intracranial pressure	**Ans: D**
Q16. Pulmonary capillary wedge pressure (PCWP) is a surrogate for: **A.** Right atrial pressure **B.** Left atrial pressure and left ventricular end-diastolic pressure **C.** Systemic vascular resistance **D.** Pulmonary vascular resistance	**Ans: B**
Q17. Pulse oximetry measures: **A.** Partial pressure of oxygen in blood **B.** Functional oxygen saturation of hemoglobin (SpO_2) **C.** Oxygen content of blood **D.** Dissolved oxygen in plasma	**Ans: B**
Q18. Pulse oximetry works on the principle of: **A.** Fluorescence **B.** Beer-Lambert law (spectrophotometry with two wavelengths of light) **C.** Piezoelectric effect **D.** Wheatstone bridge	**Ans: B**
Q19. Which of the following causes a falsely low pulse oximetry reading? **A.** Carbon monoxide poisoning **B.** Methylene blue dye injection **C.** Anemia **D.** Jaundice	**Ans: B**
Q20. Capnography measures: **A.** Oxygen saturation **B.** End-tidal carbon dioxide ($EtCO_2$) concentration **C.** Tidal volume **D.** Airway pressure	**Ans: B**
Q21. Normal end-tidal CO_2 ($EtCO_2$) value is approximately: **A.** 10–20 mmHg **B.** 35–45 mmHg **C.** 60–80 mmHg **D.** 5–10 mmHg	**Ans: B**
Q22. A sudden drop in $EtCO_2$ to zero during anesthesia most likely indicates: **A.** Bronchospasm **B.** Circuit disconnection or esophageal intubation **C.** Hyperventilation **D.** Increased cardiac output	**Ans: B**

Q23. **A gradual rise in $EtCO_2$ during anesthesia may indicate:**

A. Hyperventilation
B. Hypoventilation, increased CO_2 production (malignant hyperthermia), or rebreathing
C. Pulmonary embolism
D. Cardiac arrest

Ans: B

Q24. **Capnography works on the principle of:**

A. Mass spectrometry
B. Infrared absorption spectroscopy
C. Piezoelectric effect
D. Paramagnetic analysis

Ans: B

Q25. **Mainstream capnography differs from sidestream in that:**

A. It samples gas away from the patient circuit
B. The sensor is placed directly in the breathing circuit (at the airway)
C. It has a longer response time
D. It is less accurate

Ans: B

Q26. **Spirometry during anesthesia can measure all EXCEPT:**

A. Tidal volume
B. Minute ventilation
C. Compliance and resistance
D. Cardiac output

Ans: D

Q27. **Airway pressure monitoring helps detect all EXCEPT:**

A. Circuit disconnection
B. Bronchospasm
C. Endobronchial intubation
D. Myocardial ischemia

Ans: D

Q28. **Oxygen analyzers in the anesthetic circuit commonly use which principle?**

A. Infrared absorption
B. Paramagnetic or galvanic cell (fuel cell) principle
C. Mass spectrometry only
D. Colorimetric detection

Ans: B

Q29. **The normal core body temperature range during anesthesia should be maintained at:**

A. 34–35°C
B. 36–37.5°C
C. 38–39°C
D. 32–34°C

Ans: B

Q30. Sites for core temperature monitoring include all EXCEPT:

A. Nasopharynx
B. Distal esophagus
C. Tympanic membrane
D. Axilla

Ans: D

Q31. Perioperative hypothermia is defined as core temperature below:

A. 37°C
B. 36°C
C. 35°C
D. 34°C

Ans: B

Q32. Consequences of perioperative hypothermia include all EXCEPT:

A. Impaired coagulation
B. Increased wound infection
C. Shivering and increased oxygen consumption
D. Decreased duration of neuromuscular blockade

Ans: D

Q33. Malignant hyperthermia is suspected when temperature monitoring shows:

A. Gradual decrease in temperature
B. Rapid and unexplained rise in temperature with rising $EtCO_2$
C. Stable temperature throughout
D. Temperature of 36.5°C

Ans: B

Q34. The mechanism of redistribution hypothermia in the first hour of anesthesia is:

A. Excessive fluid administration
B. Heat redistribution from core to peripheral compartment due to vasodilation
C. Cold operating room only
D. Shivering

Ans: B

Q35. Neuromuscular monitoring is used to assess:

A. Depth of anesthesia
B. Degree of neuromuscular blockade and recovery from muscle relaxants
C. Blood pressure
D. Cerebral activity

Ans: B

Q36. The most commonly used nerve for peripheral nerve stimulation in neuromuscular monitoring is:

A. Median nerve
B. Ulnar nerve (adductor pollicis response)
C. Radial nerve
D. Femoral nerve

Ans: B

Q37. Train-of-four (TOF) stimulation consists of: **A.** Single twitch at 1 Hz **B.** Four supramaximal stimuli at 2 Hz (every 0.5 seconds) **C.** Tetanic stimulation at 50 Hz **D.** Double burst stimulation	**Ans: B**
Q38. A TOF ratio of greater than 0.9 indicates: **A.** Complete neuromuscular block **B.** Adequate recovery from neuromuscular blockade (safe to extubate) **C.** Need for more muscle relaxant **D.** Onset of block	**Ans: B**
Q39. TOF count of zero (no twitches visible) indicates: **A.** No neuromuscular block **B.** Intense (deep) neuromuscular block (>95% receptor occupation) **C.** Partial block only **D.** Equipment malfunction	**Ans: B**
Q40. Post-tetanic count (PTC) is used when: **A.** TOF shows 4 twitches **B.** TOF count is zero (intense block) to assess depth of profound block **C.** Patient is fully recovered **D.** No muscle relaxant has been given	**Ans: B**
Q41. Fade in TOF response is characteristic of: **A.** Depolarizing block (Phase I succinylcholine block) **B.** Non-depolarizing neuromuscular block **C.** No block at all **D.** Myasthenia gravis only	**Ans: B**
Q42. The Bispectral Index (BIS) monitor is used to assess: **A.** Neuromuscular blockade **B.** Depth of anesthesia/level of consciousness **C.** Blood pressure **D.** Oxygen saturation	**Ans: B**
Q43. The recommended BIS value range for adequate general anesthesia is: **A.** 80–100 **B.** 40–60 **C.** 0–20 **D.** 60–80	**Ans: B**
Q44. A BIS value of 100 indicates: **A.** Deep anesthesia **B.** Fully awake state **C.** Brain death **D.** Burst suppression	**Ans: B**

Q45. Evoked potential monitoring during anesthesia is used to assess: **A.** Blood pressure changes **B.** Integrity of neural pathways (spinal cord, brainstem, cortex) **C.** Muscle relaxation **D.** Temperature changes	**Ans: B**
Q46. Monitoring blood loss during surgery can be estimated by all EXCEPT: **A.** Weighing surgical swabs/sponges **B.** Measuring suction bottle contents **C.** Monitoring pulse oximetry alone **D.** Visual estimation and hemodynamic parameters	**Ans: C**
Q47. The formula for estimated blood loss by swab weighing is: **A.** Weight of dry swab minus weight of wet swab **B.** Weight of wet (blood-soaked) swab minus weight of dry swab (1 gram = 1 mL blood) **C.** Volume of suction divided by 2 **D.** Hematocrit multiplied by weight	**Ans: B**
Q48. Maximum allowable blood loss (MABL) depends on: **A.** Patient's height only **B.** Estimated blood volume, starting hematocrit, and minimum acceptable hematocrit **C.** Type of surgery only **D.** Duration of surgery	**Ans: B**
Q49. Urine output monitoring during surgery is a reliable indicator of: **A.** Depth of anesthesia **B.** Renal perfusion and adequacy of intravascular volume **C.** Neuromuscular blockade **D.** Blood glucose level	**Ans: B**
Q50. Which of the following is used as a point-of-care test to rapidly assess hemoglobin during surgery? **A.** CT scan **B.** HemoCue or arterial blood gas analyzer **C.** Pulse oximetry **D.** Capnography	**Ans: B**

Answer Key & Explanations

Q1. Answer: B — Observation of skin color, pupil size, and respiratory pattern

Clinical monitoring involves direct observation by the anesthesiologist including skin color, pupil size, chest movements, and peripheral perfusion without instruments.

Topic: Clinical Monitoring | Ref: Ch 6, Page 57

Q2. Answer: C — BIS value of 40–60

BIS monitoring is an instrumental (advanced) monitor, not a clinical sign. Clinical signs include stable vitals, pupil size, and absence of autonomic responses.

Topic: Clinical Monitoring | Ref: Ch 6, Page 57

Q3. Answer: B — Light anesthesia, hypoxia, or hypercarbia

Dilated pupils during anesthesia may indicate light anesthesia, hypoxia, hypercarbia, or sympathetic stimulation.

Topic: Clinical Monitoring | Ref: Ch 6, Page 57

Q4. Answer: D — Cardiac output monitoring

Minimum monitoring standards include pulse oximetry, ECG, NIBP, capnography, and temperature. Cardiac output monitoring is an advanced monitor used in select cases.

Topic: Advance Monitoring (Instrumental Monitoring) | Ref: Ch 6, Page 57

Q5. Answer: B — Arrhythmias, ischemia, and heart rate

ECG monitoring detects cardiac rhythm disturbances, myocardial ischemia (ST changes), and heart rate during anesthesia.

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

Q6. Answer: C — Lead V5

Lead V5 detects approximately 75% of intraoperative ST-segment changes indicative of left ventricular ischemia.

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

Q7. Answer: B — P waves and inferior wall ischemia

Lead II provides the best P-wave visualization for rhythm analysis and detects inferior wall ischemia.

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

Q8. Answer: B — Mean arterial pressure directly, then calculates systolic and diastolic

The oscillometric method directly measures mean arterial pressure (at maximum oscillation amplitude) and derives systolic and diastolic pressures.

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

Q9. Answer: C — Minor ambulatory surgery in healthy patients

Invasive arterial monitoring is not indicated for minor ambulatory surgery in healthy patients. It is reserved for major surgeries and hemodynamically unstable patients.

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

Q10. Answer: B — Radial artery

The radial artery is the most common site for arterial cannulation due to superficial location, good collateral circulation (ulnar artery), and accessibility.

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

Q11. Answer: B — Adequacy of collateral circulation from the ulnar artery

Allen's test assesses the adequacy of ulnar artery collateral flow to the hand before radial artery cannulation.

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

Q12. Answer: B — Junction of superior vena cava and right atrium

The CVP catheter tip should be at the junction of the SVC and right atrium for accurate pressure measurement.

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

Q13. Answer: B — 2–8 mmHg (3–11 cm H_2O)

Normal CVP ranges from 2–8 mmHg (approximately 3–11 cm H_2O).

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

Q14. Answer: B — Right heart filling pressure and intravascular volume status

CVP reflects right atrial pressure, which indicates right heart filling pressure and helps guide fluid management.

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

Q15. Answer: D — Intracranial pressure

The PA catheter measures PA pressures, PCWP, cardiac output, and mixed venous oxygen saturation. ICP requires a separate intracranial monitor.

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

Q16. Answer: B — Left atrial pressure and left ventricular end-diastolic pressure

PCWP approximates left atrial pressure and hence left ventricular preload (LVEDP) in the absence of mitral valve disease.

Topic: Cardiovascular Monitoring | Ref: Ch 6, Page 57

Q17. Answer: B — Functional oxygen saturation of hemoglobin (SpO_2)

Pulse oximetry measures the percentage of oxyhemoglobin (SpO_2) using spectrophotometric principles at two wavelengths.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

Q18. Answer: B — Beer-Lambert law (spectrophotometry with two wavelengths of light)

Pulse oximetry uses two wavelengths (660 nm red and 940 nm infrared) based on the Beer-Lambert law to determine oxygen saturation.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

Q19. Answer: B — Methylene blue dye injection

Methylene blue absorbs light at 660 nm, causing falsely low SpO_2 readings. Carbon monoxide causes falsely HIGH readings.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

Q20. Answer: B — End-tidal carbon dioxide ($EtCO_2$) concentration

Capnography continuously measures and displays the CO_2 concentration in expired gases, with $EtCO_2$ reflecting alveolar CO_2.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

Q21. Answer: B — 35–45 mmHg

Normal EtCO2 is 35–45 mmHg, which is slightly lower than arterial PaCO2 due to alveolar dead space.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

Q22. Answer: B — Circuit disconnection or esophageal intubation

A sudden drop of EtCO2 to zero indicates circuit disconnection, extubation, or esophageal intubation (no alveolar gas being sampled).

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

Q23. Answer: B — Hypoventilation, increased CO2 production (malignant hyperthermia), or rebreathing

Gradual EtCO2 rise suggests hypoventilation, increased metabolic CO2 production (e.g., malignant hyperthermia, fever), or CO2 rebreathing.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

Q24. Answer: B — Infrared absorption spectroscopy

Capnographs use infrared absorption – CO2 absorbs infrared light at a wavelength of 4.26 micrometers.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

Q25. Answer: B — The sensor is placed directly in the breathing circuit (at the airway)

Mainstream capnography places the sensor directly in the breathing circuit, providing faster response time but adding weight/dead space.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

Q26. Answer: D — Cardiac output

Intraoperative spirometry measures tidal volume, minute ventilation, flow rates, compliance, and resistance – not cardiac output.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

Q27. Answer: D — Myocardial ischemia

Airway pressure monitoring detects mechanical problems (disconnection, obstruction, bronchospasm, endobronchial intubation) but not myocardial ischemia.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

Q28. Answer: B — Paramagnetic or galvanic cell (fuel cell) principle

Oxygen analyzers use paramagnetic properties of O2 or electrochemical (galvanic/fuel cell) principles to measure FiO2.

Topic: Respiratory Monitoring | Ref: Ch 6, Page 61

Q29. Answer: B — 36–37.5°C

Core body temperature should be maintained between 36–37.5°C during anesthesia to avoid complications of hypothermia or hyperthermia.

Topic: Temperature Monitoring | Ref: Ch 6, Page 65

Q30. Answer: D — Axilla

The axilla measures peripheral (shell) temperature. Core temperature sites include nasopharynx, distal esophagus, tympanic membrane, and pulmonary artery.

Topic: Temperature Monitoring | Ref: Ch 6, Page 65

Q31. Answer: B — 36°C

Perioperative hypothermia is defined as core temperature below 36°C and is common during general anesthesia.

Topic: Temperature Monitoring | Ref: Ch 6, Page 65

Q32. Answer: D — Decreased duration of neuromuscular blockade

Hypothermia prolongs the duration of neuromuscular blockade (not decreases it). Other complications include coagulopathy, wound infection, and shivering.

Topic: Temperature Monitoring | Ref: Ch 6, Page 65

Q33. Answer: B — Rapid and unexplained rise in temperature with rising EtCO2

Malignant hyperthermia presents with rapidly rising temperature, markedly elevated EtCO2, muscle rigidity, and tachycardia.

Topic: Temperature Monitoring | Ref: Ch 6, Page 65

Q34. Answer: B — Heat redistribution from core to peripheral compartment due to vasodilation

Anesthetic-induced vasodilation causes redistribution of heat from the warm core to the cooler peripheral compartment, dropping core temperature rapidly in the first hour.

Topic: Temperature Monitoring | Ref: Ch 6, Page 65

Q35. Answer: B — Degree of neuromuscular blockade and recovery from muscle relaxants

Neuromuscular monitoring quantifies the degree of neuromuscular blockade and guides timing of reversal and extubation.

Topic: Neuromuscular Monitoring | Ref: Ch 6, Page 66

Q36. Answer: B — Ulnar nerve (adductor pollicis response)

The ulnar nerve at the wrist is most commonly stimulated, and the response of the adductor pollicis muscle is observed.

Topic: Neuromuscular Monitoring | Ref: Ch 6, Page 66

Q37. Answer: B — Four supramaximal stimuli at 2 Hz (every 0.5 seconds)

TOF consists of four supramaximal stimuli delivered at 2 Hz (0.5 second intervals), with the ratio of 4th to 1st twitch indicating block depth.

Topic: Neuromuscular Monitoring | Ref: Ch 6, Page 66

Q38. Answer: B — Adequate recovery from neuromuscular blockade (safe to extubate)

A TOF ratio ≥ 0.9 indicates adequate recovery from neuromuscular blockade and is the threshold for safe extubation.

Topic: Neuromuscular Monitoring | Ref: Ch 6, Page 66

Q39. Answer: B — Intense (deep) neuromuscular block (>95% receptor occupation)

Zero twitches on TOF indicates intense/deep block with greater than 95% of receptors occupied.

Topic: Neuromuscular Monitoring | Ref: Ch 6, Page 66

Q40. Answer: B — TOF count is zero (intense block) to assess depth of profound block

PTC is used during intense block (TOF count = 0) to quantify the depth of block by counting post-tetanic facilitation twitches.

Topic: Neuromuscular Monitoring | Ref: Ch 6, Page 66

Q41. Answer: B — Non-depolarizing neuromuscular block

Fade (progressive decrease in twitch height) is characteristic of non-depolarizing block. Phase I depolarizing block shows uniform decrease without fade.

Topic: Neuromuscular Monitoring | Ref: Ch 6, Page 66

Q42. Answer: B — Depth of anesthesia/level of consciousness

BIS is a processed EEG parameter that provides a numerical value (0–100) correlating with the depth of anesthesia/consciousness.

Topic: Central Nervous System (CNS) Monitoring | Ref: Ch 6, Page 68

Q43. Answer: B — 40–60

BIS values of 40–60 indicate adequate depth of general anesthesia with low probability of awareness.

Topic: Central Nervous System (CNS) Monitoring | Ref: Ch 6, Page 68

Q44. Answer: B — Fully awake state

BIS of 100 represents a fully awake state, while 0 represents EEG silence (isoelectric).

Topic: Central Nervous System (CNS) Monitoring | Ref: Ch 6, Page 68

Q45. Answer: B — Integrity of neural pathways (spinal cord, brainstem, cortex)

Evoked potentials (SSEP, MEP, BAEP) monitor the functional integrity of neural pathways during surgeries that risk neurological injury.

Topic: Central Nervous System (CNS) Monitoring | Ref: Ch 6, Page 68

Q46. Answer: C — Monitoring pulse oximetry alone

Pulse oximetry alone cannot estimate blood loss. Methods include swab weighing, suction measurement, visual estimation, and hemodynamic monitoring.

Topic: Monitoring Blood Loss | Ref: Ch 6, Page 69

Q47. Answer: B — Weight of wet (blood-soaked) swab minus weight of dry swab (1 gram = 1 mL blood)

Blood loss = (weight of blood-soaked swab – weight of dry swab) in grams, where 1 gram of blood ≈ 1 mL.

Topic: Monitoring Blood Loss | Ref: Ch 6, Page 69

Q48. Answer: B — Estimated blood volume, starting hematocrit, and minimum acceptable hematocrit

MABL = EBV × (Starting Hct – Minimum acceptable Hct) / Starting Hct. It depends on blood volume and hematocrit values.

Topic: Monitoring Blood Loss | Ref: Ch 6, Page 69

Q49. Answer: B — Renal perfusion and adequacy of intravascular volume

Urine output (target ≥0.5 mL/kg/hr) reflects renal perfusion and is a useful guide to the adequacy of fluid resuscitation and cardiac output.

Topic: Monitoring Blood Loss | Ref: Ch 6, Page 69

Q50. Answer: B — HemoCue or arterial blood gas analyzer

HemoCue and ABG analyzers provide rapid point-of-care hemoglobin measurement to guide transfusion decisions intraoperatively.

Topic: Monitoring Blood Loss | Ref: Ch 6, Page 69